Kathy

Data Scientist & Engineer | Melbourne, Australia

karthikapillai@gmail.com | LinkedIn | GitHub | Portfolio

Temporary Graduate Visa (Subclass 485). Available for full-time work from November 2026.

PROFILE

Master of Data Science student at Monash (2025-2026), with software engineering experience on IBM watsonx.data (Presto). Open to graduate and junior data roles in Melbourne. Focus: predictive modelling, reliable data pipelines and applied AI.

EXPERIENCE

Software Developer | IBM India Software Labs, India
August 2022 - January 2025

Worked on Presto within watsonx.data across security, performance monitoring and developer tooling. Contributed audit logging, automated security checks and caching improvements. Built Python tools to streamline quality assurance.

Coding Instructor | Junior Engineers, Melbourne
August 2025 - Present

Teach Python, introductory data science and AI concepts. Support instructor onboarding and explain complex concepts through practical lessons.

SELECTED PROJECTS

Voltwise | Python, scikit-learn, SQL, dbt, DuckDB, FastAPI, Ollama

Forecast battery capacity ten discharge cycles ahead using NASA laboratory data. Leave-one-battery-out evaluation across four cells: 0.024 Ah MAE across 560 forecast windows, 44.7% lower error than persistence. Includes 13 dbt checks and a local, read-only AI agent. Research prototype; not validated on vehicle fleets. Source code

Nexora (DataMedic) | Data pipeline diagnostics

Interactive demonstration of pipeline diagnosis and recovery using synthetic scenarios and recorded results. Makes investigation steps visible. Source code

Sensory-Aware Route Planner | Python, OpenStreetMap, GitHub Actions

Melbourne walking-route explorer using street and address data plus timestamped pedestrian-counter snapshots. Validation flags stale or incomplete observations; unavailable noise and light measurements remain unknown. Source code

EDUCATION

Master of Data Science | Monash University, Melbourne | 2025-2026
B.Tech (Honours), Computer Science Engineering | TKM College of Engineering, India

SKILLS

Data and modelling: Python, SQL, NumPy, scikit-learn, model validation.
Analytics and applications: dbt, DuckDB, semantic metrics, FastAPI, local Ollama agents.
Professional platform work: Presto, Java, audit logging, performance monitoring, Python tooling.